

Evaluating Parameter-Efficient Fine-Tuning vs. Prompt Engineering for Profile-Driven Recommendations

Ratomir Karlović ¹ , Luka Sever ¹ , Sandi Baressi Šegota ¹  and Ivan Lorencin ¹ 

¹ Faculty of Informatics, Juraj Dobrila University of Pula, Zagrebačka 30, 52100 Pula, Croatia; ratomir.karlovic@unipu.hr (R.K.), luka.sever@unipu.hr (L.S.), sandi.baressi.segota@unipu.hr (S.B.Š.), ivan.lorencin@unipu.hr (I.L.)

* Correspondence: e-mail@e-mail.com;

Abstract

Large language models (LLMs) show significant promise for personalized recommendation systems, yet identifying the most effective adaptation strategy for structured data remains a challenge. This study evaluates the performance trade-offs between parameter-efficient fine-tuning (PEFT) and in-context learning approaches for personalized item recommendation. We utilized explicit user profiles, encompassing specific likes and dislikes, alongside a comprehensive JSON-formatted item catalog. Two distinct adaptation paradigms were compared: PEFT, specifically utilizing Low-Rank Adaptation (LoRA) and Infused Adapter by Inhibiting and Amplifying Inner Activations (IA³), against robust prompt engineering strategies optimizing input instructions without altering model weights. The approaches were systematically evaluated using standard ranking metrics, including Normalized Discounted Cumulative Gain (NDCG@K) and Hit Ratio (HR@K), alongside computational efficiency measures. Our findings demonstrate that while few-shot prompt engineering establishes a competitive baseline, PEFT methods, particularly LoRA, achieve superior alignment with nuanced user preferences, yielding significant improvements in ranking accuracy. Furthermore, PEFT effectively mitigates context-window limitations, reducing inference latency compared to extensive prompt demonstrations. Consequently, we conclude that adapter-based fine-tuning provides more reliable and scalable personalized recommendations from structured catalogs than prompt engineering alone, successfully bridging raw LLM reasoning with strict domain-specific requirements.

Keywords: large language models; parameter-efficient fine-tuning; prompt engineering; recommender systems; Low-Rank Adaptation (LoRA); structured data processing; in-context learning

1. Introduction

The integration of Large Language Models (LLMs) into recommendation systems has fundamentally transformed the field, treating recommendation as a sequence generation task. However, adapting these models to dynamic user preferences presents significant challenges regarding stability and plasticity. To address the continuous evolution of users and items, recent research has explored parameter-efficient fine-tuning (PEFT) methods like Low-Rank Adaptation (LoRA); for instance, the PESO framework maintains a single evolving adapter anchored by a proximal term to dynamically capture new user behaviors while mitigating the catastrophic forgetting associated with sequential updates [1].

Transitioning LLMs into conversational recommenders introduces additional complexities, particularly concerning semantic misalignment and cumbersome training processes.

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To streamline this, the PECRS framework reformulates the problem entirely as a natural language processing task, utilizing a frozen pre-trained language model and rich textual metadata to perform both item recommendation and response generation. This unified architecture can be optimized in a single training stage using PEFT techniques, significantly reducing computational overhead [2].

While efficiency is crucial, standard PEFT methods often lack parameter-level personalization because they rely on a static, shared LoRA module for all users. The RecLoRA framework addresses this by introducing a Personalized LoRA module, which dynamically generates unique weights for each user through a soft routing mechanism guided by a conventional recommendation model [3].

Beyond parameter adaptation, prompt engineering and in-context learning offer alternative pathways for efficient LLM integration. Because routing all users through an LLM is computationally prohibitive, recent hybrid task allocation frameworks propose identifying "weak users" with sparse interaction histories and contextualizing their data into natural language instructions. This targeted approach drastically reduces token usage and latency while improving recommendation quality where traditional systems struggle [4].

Finally, applying these adaptations in complex, multi-domain industrial settings requires balancing general user behavior with domain-specific data sparsity. Frameworks such as MLoRA achieve this by combining a shared backbone, which learns generalizable information across domains, with specialized low-rank adaptive modules tailored to capture distinctive, domain-specific features [5].

2. Background and Related Work

The integration of Large Language Models (LLMs) into recommender systems has been systematically categorized by Elmoghazy et al., who established a taxonomy based on the LLM's specific role, learning paradigms, and overarching system architecture. Their comparative study emphasizes that while LLMs can act as textual feature encoders or direct recommendation generators, they face persistent challenges such as context window constraints, computational latency, and a disparity between language objectives and recommendation tasks [6]. Further structuring this domain, Bao et al. surveyed the field by dividing architectures into Discriminative and Generative paradigms. Their research details how generative models provide impressive zero-shot capabilities and natural language explainability, despite struggling with context length limits and inherent biases [7]. Additionally, Liu et al. explored language modeling paradigm adaptations, contrasting traditional pre-training with parameter-efficient prompt learning to overcome data sparsity, while noting ongoing issues with factual consistency and user privacy [8].

Transitioning from theoretical surveys to empirical frameworks, Xu et al. proposed LLMRec to systematically evaluate the performance of LLMs functioning directly as recommender systems. Their extensive empirical analysis demonstrated that fine-tuning open-source models via parameter-efficient methods like Low-Rank Adaptation (LoRA) yields superior domain adaptation compared to zero-shot approaches, although targeted reasoning prompts like Chain-of-Thought remain crucial for optimizing prediction accuracy [9].

Focusing specifically on the advancement of parameter-efficient methods, Zhu et al. addressed the inefficiency of static LoRA matrices by introducing RecLoRA. This framework dynamically aggregates multiple independent meta-LoRA weights to create a unique, personalized module for each user, successfully balancing recommendation effectiveness and processing time [3]. Building further on personalized adaptation, Kim et al. identified that standard PEFT foundation models often generate user-blind item embeddings. To

resolve this, they proposed PerPEFT, a framework that groups users by similar interests and assigns a distinct, dedicated PEFT module to each demographic. By employing group-specific hard negative sampling, their architecture dynamically captures fine-grained item characteristics, proving highly adaptable to various structures like LoRA and IA³ with minimal parameter overhead [10]. Addressing the dynamic nature of user profiles over time, Zeng et al. tackled the issue of shifting preferences and catastrophic forgetting associated with global fine-tuning. They introduced the Region-Aware Incremental Editing framework, which clusters historical interactions into distinct knowledge regions. When new behaviors emerge, a confidence-aware routing mechanism triggers localized updates to a specialized LoRA module dedicated solely to the affected region, effectively balancing the assimilation of novel interests with the retention of stable historical behaviors [11].

In domain-specific applications, Guo et al. utilized Quantized LoRA within the TIGMI framework to fine-tune an LLM for cultural tourism. This approach reduced computational overhead by 60 percent while successfully integrating Retrieval-Augmented Generation (RAG) for real-time multimodal interactions [12]. Similarly, Choi et al. applied LoRA to the upper layers of a BERT-based model to enhance a RAG framework for autonomous driving, effectively reducing trainable parameters and improving the retrieval of complex legal texts [13]. Furthermore, Gao et al. demonstrated the scalability of fine-tuned LLMs by developing a multi-dimensional framework to automatically assess the quality and reliability of massive volumes of tourist-generated contents across digital platforms [14].

Conversely, prompt engineering offers an effective alternative that completely avoids altering model weights. Nikolaychuk and Poddubniy applied prompt engineering to the Saiga2 model to automatically extract and classify user sentiments from regional tourism data, successfully achieving high alignment with human expert evaluations [15]. To resolve the complex challenges of intent recognition and factual hallucinations, Huang et al. proposed a multi-layer prompt engineering architecture that decomposes user queries into sequential subtasks. By integrating this step-by-step reasoning strategy with real-time external APIs, their system drastically reduced hallucination rates, proving particularly effective for lightweight models in travel planning applications [16].

3. Methodology

Materials and Methods should be described with sufficient details to allow others to replicate and build on published results. Please note that publication of your manuscript implicates that you must make all materials, data, computer code, and protocols associated with the publication available to readers. Please disclose at the submission stage any restrictions on the availability of materials or information. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited.

Research manuscripts reporting large datasets that are deposited in a publicly available database should specify where the data have been deposited and provide the relevant accession numbers. If the accession numbers have not yet been obtained at the time of submission, please state that they will be provided during review. They must be provided prior to publication.

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4. Results and discussion

5. Limitations and Future Work

6. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

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Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA ³	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
F1	F1 Score
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

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